

Matching Methods for Multilevel Data in Education Research: A Practical Guide and Illustration of Approaches in R

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Agenda

1. Foundational concepts (a quick reminder)
2. Typology for matching in multilevel settings
3. Practice guide overview
4. Practice guide data and R packages (Chp 4)
-- 10 minute break --
5. Practice guide walkthrough: MAID (Chp 5)
6. Practice guide walkthrough: CAD (Chp 6)
7. Q&A



Motivation

- Propensity score matching is a common tool for estimating the effects of practices, programs, and policies
- Most guidance regarding the use of propensity scores overlooks important complications that arise from the multilevel nature of education data
- Some research addresses matching in multilevel settings, but limited in scope and no synthesis of that work for education researchers

Provide applied researchers with a more comprehensive understanding of, and framework for, the use of propensity score matching in education research

Foundational Concepts (A Quick Reminder)

Potential Outcomes Framework

- Individual causal effect (δ_i) is defined as the difference between two potential outcomes:

$$\delta_i = Y_i(1) - Y_i(0)$$

- Fundamental problem of causal inference

- Can estimate the average causal effect based on the observed data:

$$\hat{\delta} = E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$$

- Will be biased unless certain assumptions are met

Key Assumptions for Causal Inference

- Stable-unit-treatment-value assumption (SUTVA)
 - The **no interference assumption**: the potential outcomes for a given unit do not depend on the treatment received by another unit
 - The **consistency assumption**: the treatment condition a unit is assigned is invariant across units (i.e., no different versions of the treatment)
- Strong ignorability of treatment assignment
 - The **ignorable treatment assignment assumption**: selection into, or assignment to, the treatment condition is not associated with the potential outcomes or is conditionally independent
 - The **positivity assumption**: all individuals in the analysis have a non-zero probability of receiving the treatment

Matching Methods

Matching methods aim to “equate (or ‘balance’) the distribution of covariates in the treated and control groups” (Stuart, 2010, p. 1)

The immediate goal of matching is to improve balance, the degree to which the treatment and control covariate distributions resemble each other ... without losing too many observations in the process. The result of this process, when done appropriately, is considerably less model dependence, reduced potential for bias, less variance, and as a result lower mean squared error.

Ho et al., 2007, pp. 215-216

Matching Methods

A)

Original Sample



D)

Full Matching



B)

1:1 Pair Matching with Caliper



E)

Weighting by the Odds



Image Source: Greifer, N., & Stuart, E. A. (2021). Matching Methods for Confounder Adjustment: An Addition to the Epidemiologist's Toolbox. *Epidemiologic Reviews*, 43(1), 118–129. <https://doi.org/10.1093/epirev/mxab003>

The Propensity Score

- Propensity score: probability of treatment selection conditional on a set of covariates ($\mathbf{X}$):

$$p(\mathbf{X}) = \Pr(D = 1 \mid \mathbf{X})$$

- Logistic regression is the most common estimation approach:

$$\text{logit}(\hat{p}(\mathbf{X})) = \beta_0 + \boldsymbol{\beta}_k \mathbf{X}_k$$

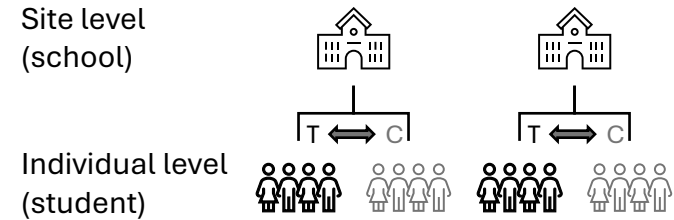
- Machine learning methods are becoming a popular way to estimate the propensity score

Typology for Matching in Multilevel Settings

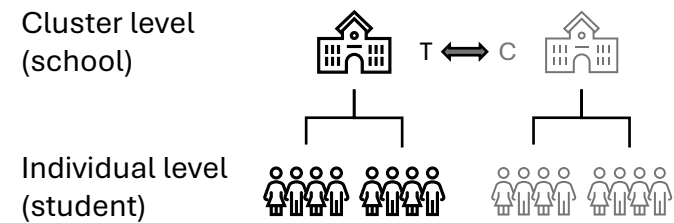
Multilevel Designs

- **individuals** = level at which the outcome measure is defined (e.g., students)
- **clusters** = grouping level at which treatment assignment occurs when assignment is not at the individual level (e.g., teachers or schools)
- **sites** = level within which individuals and/or clusters are nested (e.g., schools for a study in which students within schools are assigned to an intervention)

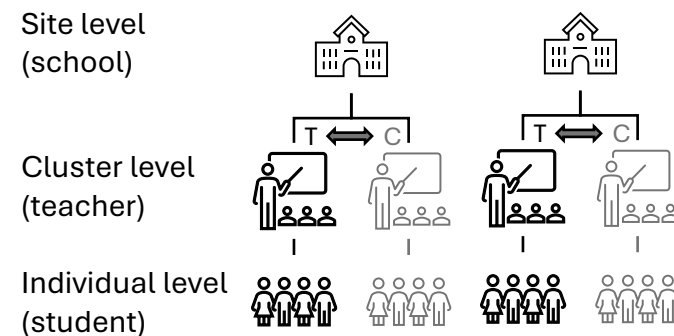
(a) Multisite Individual Assignment Design (MIAD)



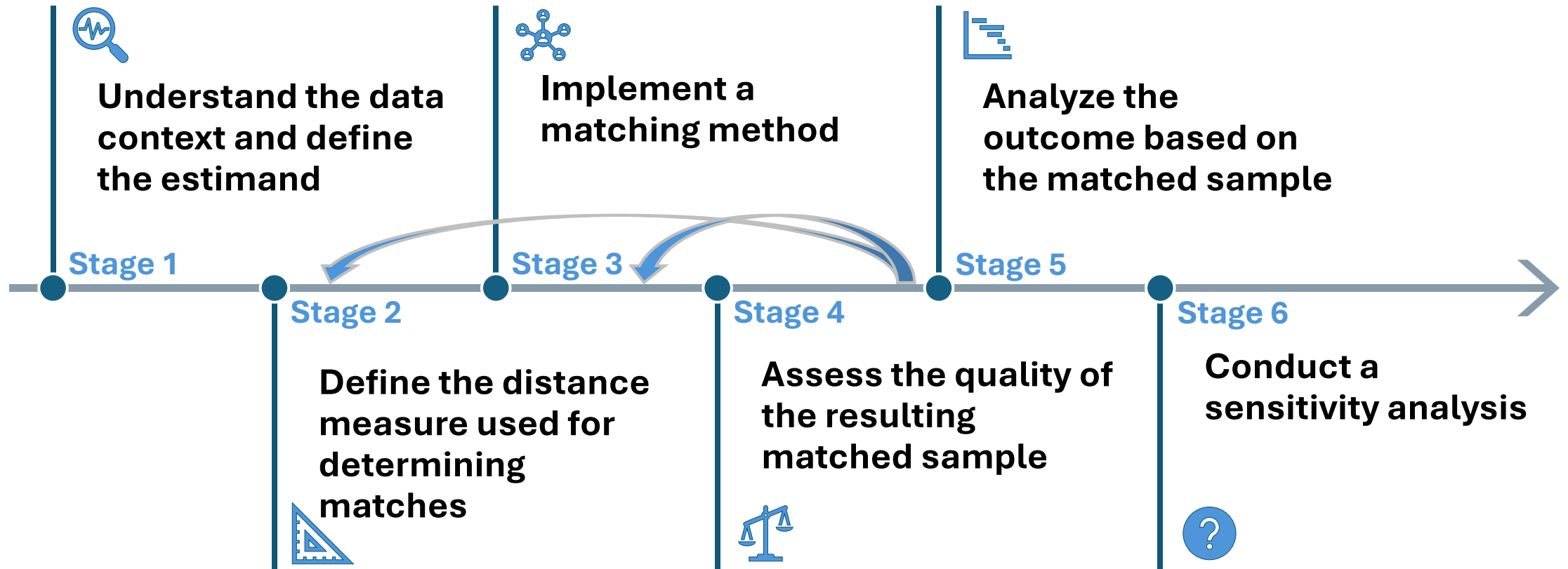
(b) Cluster Assignment Design (CAD)



(c) Multisite Cluster Assignment Design (MCAD)



Six-Stage Matching Analysis Workflow



Workflow adapted from Stuart (2010), Leiti (2017), and Zhao et al. (2021)

Stage 1: Understanding the Data Context

- Do the within-group and/or group-level sample sizes raise concerns about the feasibility of some matching approaches?
- To what extent does the proportion of treatment individuals differ across sites? (MIAD and MCAD only)
- In what ways are the treatment individuals and/or clusters different from the comparison pool of individuals and/or clusters? Are there concerns about common support?

Stage 1: Define Your Estimand of Interest

- Build off the work on estimands for multisite (Miratrix et al., 2021; Raudenbush & Schwartz, 2020) and cluster (Kahan et al., 2023) randomized designs
- Between-site effect variance may also be of interest for MIAD and MCAD

Design	Average Treatment Effect Estimand		
	Individual level	Cluster level	Site level
Multisite individual assignment design (MIAD)	$\frac{1}{N} \sum_1^N \delta_{ik}$	N/A	$\frac{1}{K} \sum_1^K \sum_1^{N_k} \frac{\delta_{ik}}{N_k}$
Cluster assignment design (CAD)	$\frac{1}{N} \sum_1^N \delta_{ij}$	$\frac{1}{J} \sum_1^J \sum_1^{N_j} \frac{\delta_{ij}}{N_j}$	N/A
Multisite cluster assignment design (MCAD)	$\frac{1}{N} \sum_1^N \delta_{ijk}$	$\frac{1}{J} \sum_1^J \sum_1^{N_{jk}} \frac{\delta_{ijk}}{N_{jk}}$	Cluster-based: $\frac{1}{K} \sum_1^K \left(\frac{1}{J_k} \sum_1^{J_k} \sum_1^{N_{jk}} \frac{\delta_{ijk}}{N_{jk}} \right)$ Individual-based: $\frac{1}{K} \sum_1^K \sum_1^{N_k} \frac{\delta_{ik}}{N_k}$

Stage 2: Define the Distance Measure

- Consider the assignment mechanism and relative importance of site/cluster factors versus individual-level factors
 - Is the site/cluster nesting “incidental” or an important feature of the design?
 - Examine the intraclass correlation (ICC) for pre-treatment factors
- Consider how the propensity score estimation approach will align with the intended design and estimand
 - MIAD: prioritize individual-level vs. site-level confounding? Individual-average vs. site-average estimand?
 - CAD: define the propensity score at the individual level, cluster level, or some combination of the two?

Stage 2: Define the Distance Measure

Method	Site effects	Unobserved site confounders	Slopes vary across sites	Parameter burden	Recommended context
Single-level logistic	None (can include observed site-level covariates)	Not controlled	No	Low	Sites are incidental and $ICC \approx 0$
Fixed effects (intercepts only)	Site indicators	Controlled via intercepts	No	Moderate (grows with sites)	Unobserved site confounding; moderate number of sites
Fixed effects (fully interacted) or site-specific single-level logistic	Site indicators × covariates	Controlled	Yes	High	Site-specific effects; sufficient within-site N
Random intercepts (RI)	Random intercepts	Partially controlled	No	Moderate	Many sites; overall effect is the target
Random intercepts and slopes (RIS)	Random intercepts + slopes	Partially controlled	Yes	Moderate–High	Site-specific effects; selection varies across sites
Machine learning (GBM/BART)	Fixed or random	Depends on specification	Flexible	Data-driven	High-dimensional covariates; functional form unknown

Stage 3: Implement a Matching Method

- Want an approach that will result in a **focal local comparison group**: “in the same locale as the treatment group and focused on persons with the same kinds of characteristics as those in the treatment group” (Shadish and Cook, 2009)
- May have to make trade-offs based on sample size constraints and prioritizing balance on focal vs. local characteristics
- Consider how the matching approach will align with the intended design and estimand

Stage 3: Implement a Matching Method

Matching approach	Preferred design	Preferred estimand	Matching priority	Recommended context	Main Limitation
Global	MIAD, CAD	Individual-average ATT	Focal	Little concern about site- or cluster-level confounding	Ignores nested structure of the data
Within-site	MIAD	Site-specific ATT and between-site ATT variance	Local	Concerned about unobserved site-level confounding	Not effective when within-site sample sizes are small and/or common support is a concern
Partially-pooled	MIAD	Individual-average ATT	Balance focal and local	Some concern about site-level confounding and within-site matching not feasible	Grouping sites may not account for unobserved site confounding; can estimate group-specific treatment effects but not site-specific effects
Preferential within-site	MIAD	Site-specific ATT and between-site ATT variance	Balance focal and local	Concerned about unobserved site-level confounding but within-site matching is limited in some sites	Mix of within- and between-site matching complicates interpretation
Cluster-level global	CAD	Cluster-average ATT	Local	Cluster-level factors drive treatment selection	Not effective when cluster sample sizes are small and/or cluster-level common support is a concern
Sequential	CAD	Cluster-average or Individual-average ATT	Balance focal and local	Distribution of individual-level factors differs across clusters or possible confounding due to individual-level selection	May not provide optimal covariate balance
Network flow optimization	CAD	Cluster-average or Individual-average ATT	Balance focal and local	Distribution of individual-level factors differs across clusters or possible confounding due to individual-level selection	May not be effective when cluster sample sizes are small

Stage 4: Assess the Quality of the Match

- Three common metrics to assess quality
 - Proportion of the full sample retained in the matched sample
 - Covariate standardized mean difference (SMD)
 - Covariate variance ratio
- Assessment of covariate balance should align with the intended design and estimand
 - For MIAD and MCAD, examine global and within-site metrics
 - For CAD, examine individual-level and cluster-level metrics
- If the quality is not acceptable, can revisit decisions in Stage 2 and Stage 3 and adjust the matching approach

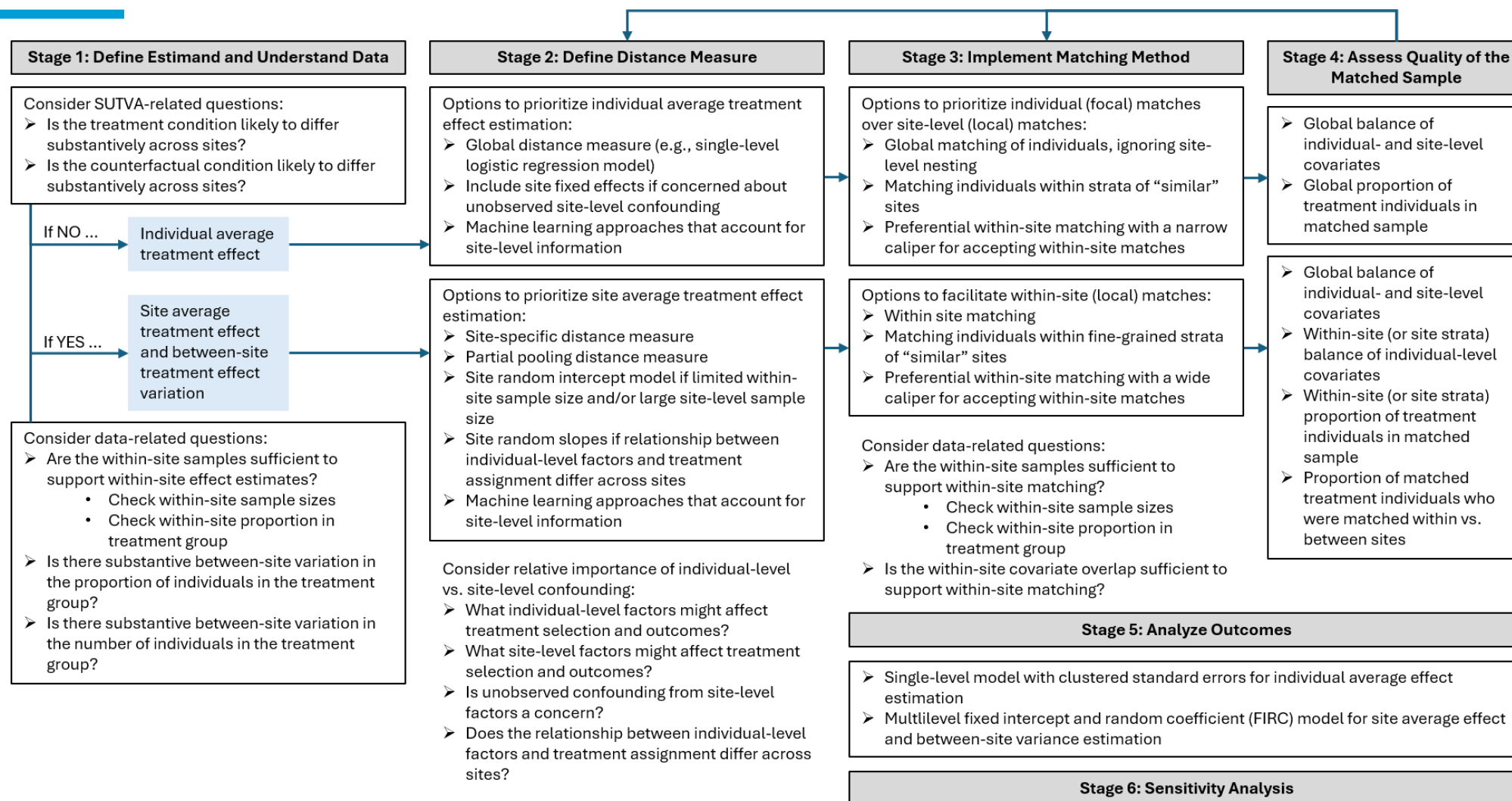
Stage 5: Analyze the Outcome

- Outcome model should align with the intended design and estimand
- Use a doubly-robust approach: matched sample + outcome model with covariate adjustment
- G-computation to estimate marginal and/or conditional average effects

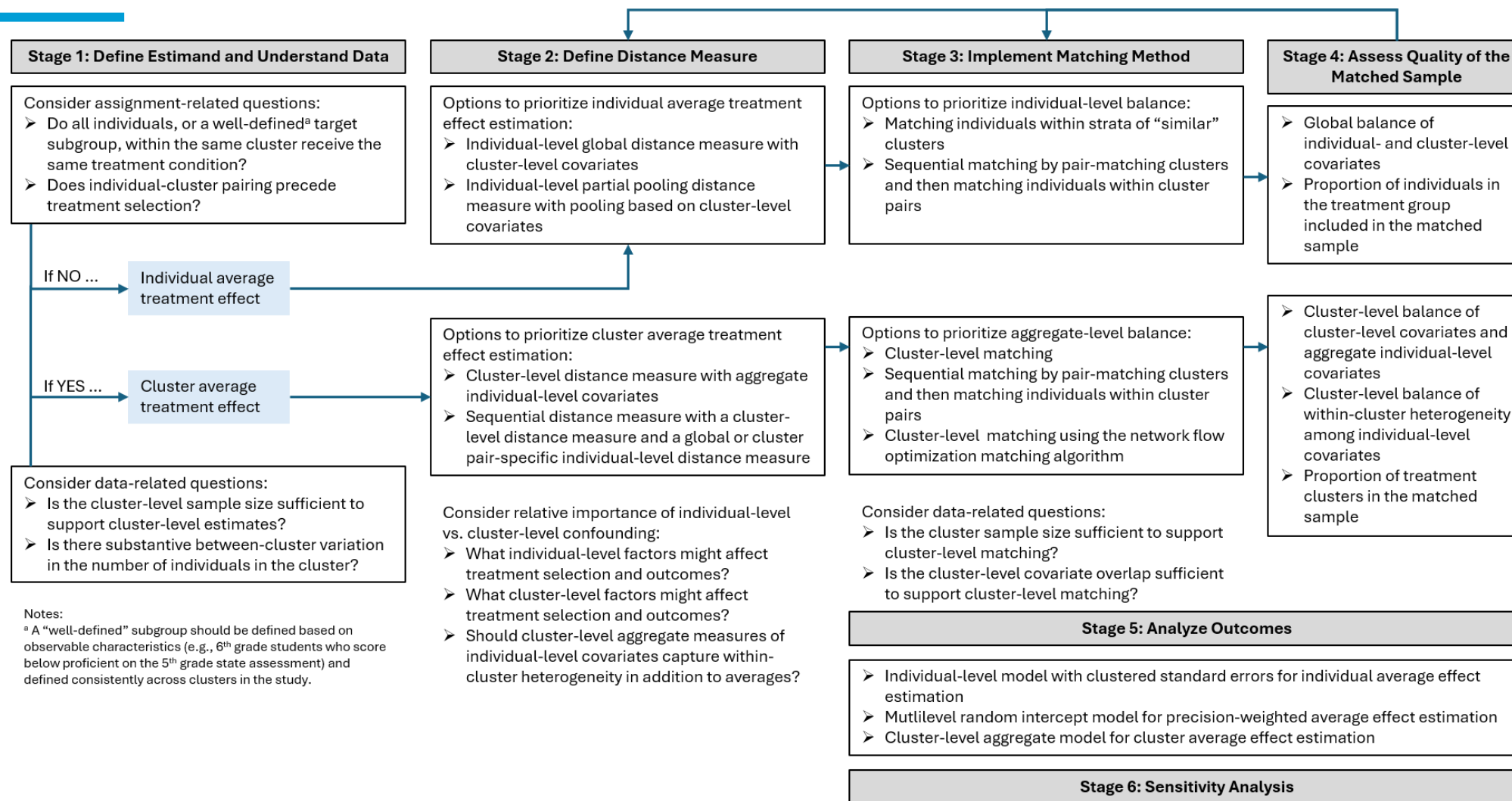
Stage 6: Conduct a Sensitivity Analysis

- How sensitive are the estimated effects and inferences to potential unobserved confounding (selection bias)?
- Can quantify how correlated the omitted variable would have to be with the outcome and treatment assignment to invalidate the causal conclusion (Cinelli & Hazlett, 2020; Frank, 2000)
- The model-estimated ICC can inform the plausible severity of confounding at individual vs. site/cluster level

Matching Guidance for a Multisite Individual Assignment Design



Matching Guidance for a Cluster Assignment Design



Practice Guide Overview

Preview of the Practice Guide



- A **primer on propensity score matching in multilevel settings** that includes an overview of propensity score matching (Chapter 2) and a discussion of the conceptual and practical issues researchers should consider when extending matching methods to a multilevel educational study (Chapter 3).
- A **handbook for education researchers** that walks through the different stages of a propensity score analysis for different multilevel designs, including worked examples and guidance on the critical analytic decisions researchers must make in each stage of the propensity score analysis (Chapters 4 - 7).
- A **compendium of supplemental resources**

Questions?

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SREE 2026 Multilevel Matching
Workshop Feedback

